

On the Erdős-Straus Conjecture: Algebraic Analysis and Modular Decomposition

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Abstract

This document presents a rigorous analysis and a structural decomposition of the Erdős-Straus conjecture. We expose the axiomatic definitions, review relevant contextual literature, isolate key lemmas concerning specific congruence classes, and propose a formalization architecture adapted for a Lean 4 interactive theorem prover.

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1 Axiomatic Definitions and Context

Definition 1.1. For any integer $n \in \mathbb{Z}$ with $n \geq 2$, the Erdős-Straus equation is defined as the Diophantine equation:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

where the variables $x, y, z \in \mathbb{Z}_{>0}$ are strictly positive integers.

1.1 Contextual Literature Research

The problem was formulated by Paul Erdős and Ernst G. Straus in 1948. Previous works on Diophantine equations of this type utilize local and global analytic principles. A notable approach is the use of modular reduction and polynomial parametrization to solve specific congruence classes. As demonstrated in the paper "*A Complete Congruence System for the Erdos-Straus Conjecture*" (available on ArXiv), analyzing the structure of solutions through decomposition into sums of fractions with identically sized numerators yields significant insights into the conjecture.

2 Proof Strategy and Lemma Isolation

The selected approach is to partition the space of integers n according to their congruence class modulo a highly composite integer, typically 4 for base cases. We break the conjecture down into specific intermediate sub-problems.

Lemma 2.1. For $n = 4k + 3$ where $k \in \mathbb{Z}_{\geq 0}$, the Erdős-Straus equation admits a solution.

Proof. Let $n = 4k + 3$ for some $k \in \mathbb{Z}_{\geq 0}$. We explicitly construct a solution by setting $x = k + 1$, $y = n(k + 1) + 1$, and $z = n(k + 1)(n(k + 1) + 1)$. First, we substitute $x = k + 1$ into the equation. We aim to show that:

$$\frac{4}{n} - \frac{1}{k + 1} = \frac{1}{y} + \frac{1}{z}$$

We evaluate the left-hand side with a common denominator $n(k + 1)$:

$$\begin{aligned} \frac{4}{n} - \frac{1}{k + 1} &= \frac{4(k + 1) - n}{n(k + 1)} \\ &= \frac{4k + 4 - (4k + 3)}{n(k + 1)} \\ &= \frac{1}{n(k + 1)} \end{aligned}$$

Now, we express $\frac{1}{n(k+1)}$ as a sum of two unit fractions using the standard Egyptian fraction identity $\frac{1}{A} = \frac{1}{A+1} + \frac{1}{A(A+1)}$. Setting $A = n(k + 1)$, we obtain:

$$\frac{1}{n(k + 1)} = \frac{1}{n(k + 1) + 1} + \frac{1}{n(k + 1)(n(k + 1) + 1)}$$

By identifying the terms, we set:

$$\begin{aligned} y &= n(k+1) + 1 \\ z &= n(k+1)(n(k+1) + 1) \end{aligned}$$

Since $n \geq 3$ and $k \geq 0$, it is clear that $x > 0$, $y > 0$, and $z > 0$. Therefore, $(x, y, z) = (k+1, n(k+1) + 1, n(k+1)(n(k+1) + 1))$ is a valid solution in positive integers. $\square$

3 Architecture for Autoformalization

The formalization of the Erdős-Straus conjecture requires structuring the statement and the search space decomposition in Lean 4.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Tactic.Omega
import Mathlib.Tactic.Ring

-- Definition axiomatique de la propriete d'Erdos-Straus
def SatisfiesErdosStraus (n : Nat) : Prop :=
  Exists (fun x => Exists (fun y => Exists (fun z => x > 0 /\ y > 0
    /\ z > 0 /\ 4 * x * y * z = n * (y * z + x * z + x * y))))

-- Lemme mod 4 = 3
lemma erdos_straus_mod_4_3 (k : Nat) : SatisfiesErdosStraus (4 * k
  + 3) := by
  -- Proof implementation pending
  sorry

-- Theoreme principal
theorem erdos_straus_conjecture (n : Nat) (hn : n >= 2) :
  SatisfiesErdosStraus n := by
  -- Proof implementation pending
  sorry
```